

Session 3

Q1. Create a SQL table called Restaurants with columns: id (INT, auto-increment), name (VARCHAR), cuisine (VARCHAR), rating (DECIMAL), and city (VARCHAR). Insert 5 sample restaurants into the table, each with a different cuisine and rating.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```
48 CREATE TABLE restaurants(  
49     id INTEGER PRIMARY KEY AUTO_INCREMENT,  
50     name VARCHAR(100),  
51     cuisine VARCHAR(100),  
52     rating DECIMAL,  
53     city VARCHAR(100));  
54 INSERT INTO restaurants(name, cuisine, rating, city)VALUES(  
55     'Hot&More', 'Butter Chicken', 5, 'Rajkot'),
```

The Result Grid shows the following data:

#	id	name	cuisine	rating	city
1	1	Hot&More	Butter Chicken	5	Rajkot
2	2	Ajmeri	Chicken 65	4	Diu
3	3	Rainbow	Biryani	4	Daman
4	4	Krishna	Lollipop	5	Diu
5	5	Maheshwari	Chicken Masala	3	Surat

The Output tab shows the following actions:

#	Time	Action	Message	Duration / Fetch
17	16:01:36	DROP TABLE restaurants	0 row(s) affected	0.031 sec
18	16:01:39	DROP TABLE restaurants	Error Code: 1051. Unknown table 'instancione.restaurants'	0.000 sec
19	16:07:25	CREATE TABLE playlists(playlist_id INTEGER PRIMARY KEY AUTO_INCREMENT, title VARCHAR(50), ...	0 row(s) affected	0.031 sec
20	16:07:38	SELECT * FROM playlists LIMIT 0. 1000	0 row(s) returned	0.000 sec / 0.000 sec
21	16:24:18	CREATE TABLE restaurants(id INTEGER PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100), ...	0 row(s) affected	0.032 sec

The screenshot shows the MySQL Workbench interface with the following code in the SQL editor:

```
54 INSERT INTO restaurants(name, cuisine, rating, city)VALUES(  
55     'Hot&More', 'Butter Chicken', 5, 'Rajkot'),  
56     ('Ajmeri', 'Chicken 65', 4, 'Diu'),  
57     ('Rainbow', 'Biryani', 4, 'Daman'),  
58     ('Krishna', 'Lollipop', 5, 'Diu'),  
59     ('Maheshwari', 'Chicken Masala', 3, 'Surat');  
60 SELECT * FROM restaurants;  
61
```

The Result Grid shows the same data as the previous screenshot:

#	id	name	cuisine	rating	city
1	1	Hot&More	Butter Chicken	5	Rajkot
2	2	Ajmeri	Chicken 65	4	Diu
3	3	Rainbow	Biryani	4	Daman
4	4	Krishna	Lollipop	5	Diu
5	5	Maheshwari	Chicken Masala	3	Surat

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20	16:07:38	SELECT * FROM playlists LIMIT 0. 1000	0 row(s) returned	0.000 sec / 0.000 sec
21	16:24:18	CREATE TABLE restaurants(id INTEGER PRIMARY KEY AUTO_INCREMENT, name VARCHAR(100), ...	0 row(s) affected	0.032 sec

Q2. Write a SQL UPDATE statement to change the rating of the restaurant named 'Spice Villa' to 4.7 in the Restaurants table.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
59 ('Spice Villa','Chicken Masala',3,'Surat');
60
61 SELECT * FROM restaurants;
62
63 UPDATE restaurants SET rating=4 where id=5;
64
```

The Result Grid shows the output of the SELECT query:

	id	name	cuisine	rating	city
1	HotMore	Butter Chicken	5	Rajkot	
2	Alpeni	Chicken 65	4	Diu	
3	Rainbow	Biryani	4	Daman	
4	Krishna	Lollipop	5	Diu	
5	Spice Villa	Chicken Masala	4	Surat	

The Output tab shows the execution of the UPDATE query:

#	Time	Action	Message	Duration / Fetch
28	16:37:02	SELECT * FROM restaurants LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
29	16:37:58	UPDATE restaurants SET rating=4.7 where id=5	1 row(s) affected, 1 warning(s): 1265 Data truncated for column 'rating' at row 1 Rows matched: 1 Changed: 1 ...	0.000 sec
30	16:38:17	SELECT * FROM restaurants LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
31	16:38:29	UPDATE restaurants SET rating=4 where id=5	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.047 sec
32	16:38:33	SELECT * FROM restaurants LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

Q3. Delete all restaurants from the Restaurants table where the rating is less than 3.5 using a SQL DELETE statement.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following queries:

```
52 drop table restaurants;
53
54 UPDATE restaurants SET rating=4 where id=5;
55
56 DELETE FROM restaurants where rating <= 3.5;
```

The Result Grid shows the output of the DELETE query:

	id	name	cuisine	rating	city
2	Alpeni	Chicken 65	4	Diu	
3	Rainbow	Biryani	4	Daman	
4	Krishna	Lollipop	5	Diu	

The Output tab shows the execution of the DELETE query:

#	Time	Action	Message	Duration / Fetch
46	16:44:25	DELETE restaurants where rating >= 3.5	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
47	16:44:37	DELETE restaurants where rating <= 3.5	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
48	16:44:43	DELETE restaurants where rating <= 3	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to your MySQL se...	0.000 sec
49	16:45:16	DELETE FROM restaurants where rating <= 3.5	2 row(s) affected	0.000 sec
50	16:45:20	SELECT * FROM restaurants LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Q4. Use a SELECT query with WHERE and LIMIT to fetch the top 2 restaurants in the city 'Ahmedabad' with a rating above 4.0 from the Restaurants table. Hint: Combine WHERE for city and rating, and LIMIT to restrict the number of results.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```
58 • SELECT *
59 FROM restaurants
70 WHERE city = 'Ahmedabad'
71 AND rating > 4
72 LIMIT 2;
```

The Results tab shows the output of the query as a table with 5 columns: id, name, cuisine, rating, and city. The results are as follows:

id	name	cuisine	rating	city
2	BBQ	Chicken	5	Ahmedabad

The bottom panel shows the Query History with the following entries:

#	Time	Action	Message	Duration / Fetch
55	16:52:13	SELECT * FROM restaurants LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec
56	16:52:17	SELECT * FROM restaurants WHERE city = 'Ahmedabad' AND rating > 4 LIMIT 2	0 row(s) returned	0.000 sec / 0.000 sec
57	16:53:07	update restaurants set name='BBQ', city='Ahmedabad', rating=5 where id=2	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.063 sec
58	16:53:10	SELECT * FROM restaurants LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
59	16:53:20	SELECT * FROM restaurants WHERE city = 'Ahmedabad' AND rating > 4 LIMIT 2	1 row(s) returned	0.000 sec / 0.000 sec